

Pandas Missing Values

Extra Questions Part-2

```
data = {
    "Student": ["Aman", None, "Ranu", "Neha", "Tina", None],
    "City": ["Delhi", "Mumbai", None, "Surat", None, "Pune"],
    "Age": [18, 19, None, 20, None, 21],
    "Python": [85, None, 70, 90, None, 75],
    "Django": [88, 80, None, None, 92, 85],
    "Attendance": [None, 80, 95, None, 85, 90]
}
```

Task 1 — Show Which Values Are Missing

Use:

```
df.isnull()
```

Task 2 — Count Missing Values Column-wise

Use:

```
df.isnull().sum()
```

Task 3 — Count Total Missing Values

Count total NaN in entire dataset.

Task 4 — Remove All Rows Containing NaN

Use:

```
df.dropna()
```

Task 5 — Remove All Columns Containing NaN

Use:

```
df.dropna(axis=1)
```

Task 6 — Fill Missing Values With 0

Use:

```
df.fillna(0)
```

Task 7 — Fill Age Column With Mean

Replace missing Age with:

```
df["Age"].fillna(df["Age"].mean())
```

Task 8 — Fill Python Marks With Median

Use:

```
df["Python"].fillna(df["Python"].median())
```

Task 9 — Fill Missing Student Names With "Unknown"

Use:

```
df["Student"].fillna("Unknown")
```

Task 10 — Forward Fill

Use:

```
df.fillna(method='ffill')
```

Task 11 — Backward Fill

Use:

```
df.fillna(method='bfill')
```

Task 12 — Conditional Fill (Attendance)

If **Attendance** is missing → fill 75

Task 13 — Conditional Fill (City)

If City missing → fill with "Unknown City"

Task 14 — Different Rules for Different Column